

Expected-Information-Gain-Guided Bayesian Calibration of Magnetic-Core Loss and Permeability Models for Power Converters

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Abstract

Magnetic-core models used in power-converter design are commonly fitted to nominal material curves despite limited operating-point coverage and material variation. We present a Bayesian framework that calibrates Steinmetz core-loss and Cole–Cole complex-permeability models and uses expected information gain (EIG) to select measurements. Across five matched-model synthetic data sets, per-parameter median absolute relative errors of the posterior medians ranged from 0.27% to 5.85%, and the known values fell within 28/30 equal-tailed 90% model-conditional posterior credible intervals. In a separate 30-seed paired acquisition study, raw EIG reached a prespecified two-target latent mean-response precision gate using 4–5 measurements; a fixed channel-balanced grid traversal required 9–9, giving a mean paired reduction of 44.8%. The paired count-difference bootstrap 95% interval was 4.00–4.10 measurements, with 100.0% paired wins and a 0.0% raw-EIG gate-failure rate. For accepted measured records, in-sample relative root-mean-square errors ranged from 8.79% to 18.21% for core loss, from 6.89% to 9.33% for μ' , and from 36.77% to 52.42% for μ'' . The larger μ'' residuals identify a limitation of the tested single-pole permeability model. These results support uncertainty-aware calibration and model-conditional measurement planning within the tested models and operating ranges; they do not establish an optimal laboratory plan.

Keywords: Bayesian calibration; magnetic core; core loss; complex permeability; optimal experimental design; expected information gain.

I. INTRODUCTION

Power-converter efficiency, thermal margin, and magnetic-component sizing depend on models for volumetric core loss and complex permeability. Vendor curves provide essential design information, but cover a finite set of frequencies, flux densities, and temperatures and normally do not provide a joint parameter distribution. Fitting one nominal curve can therefore hide parameter non-identifiability and model discrepancy.

Bayesian calibration represents uncertainty in model parameters conditional on the selected model, observations, and prior. It also supports Bayesian optimal experimental design: a candidate measurement can be ranked by its expected reduction of posterior uncertainty [1, 2]. This is useful when characterization time is limited and dense uniform sweeps are costly. It also makes the measurement objective explicit: the selected point should reduce uncertainty in the quantities represented by the model, not merely occupy an unmeasured location in a frequency–flux grid.

This work contributes (1) a reproducible posterior for-

mulation for Steinmetz and Cole–Cole parameters, (2) a Fisher-spectrum diagnosis of locally constrained parameter combinations, and (3) a paired comparison of EIG acquisition with a fixed channel-balanced grid traversal. These elements address distinct questions: matched-model recovery checks inference against known synthetic truth, Fisher analysis assesses local identifiability under the specified experiment, and measured-data residuals assess model adequacy. Success in the first two does not establish the third.

The remainder of the paper reviews magnetic and Bayesian design literature, defines the forward and statistical models, documents the reproducible protocol, and then reports synthetic and measured-data results from one pre-specified evaluation.

II. RELATED WORK

2.1 Magnetic loss and permeability models

The Steinmetz power law remains a compact engineering description for sinusoidal loss, while loss-separation and hysteresis models provide more physical decompositions [3, 4, 5]. Converter waveforms motivated modified and generalized Steinmetz relations that integrate the instantaneous flux-density trajectory or use waveform correction factors [6, 7, 8]. Subsequent work improved practical loss calculation and reviewed the physical contributions to loss in soft ferrites [9, 10]. Textbook treatments emphasize that material law, geometry, winding, and thermal assumptions must remain distinct in component design [11, 12].

Frequency-dispersive permeability is commonly described through relaxation or resonance spectra. The Cole–Cole law introduced a broadened relaxation distribution [13]; applying its algebraic form to permeability is an engineering adaptation rather than a claim that ferrite magnetization is a dielectric process. Measurements of Mn–Zn and Ni–Zn ferrites show that domain-wall motion, rotational processes, resonance, and relaxation can all shape μ' and μ'' [14]. The single-pole model used here is therefore intentionally diagnostic: a large loss-component residual is evidence for missing dynamics, not a parameter-estimation failure alone.

Open magnetic datasets now permit reproducible comparisons beyond individual vendor plots. MagNet provides systematically measured core-loss data across materials and excitations [15], while the LEA material database publishes machine-readable permeability and loss records with source metadata [16]. Public data improve traceability but do not substitute for a controlled multi-lot characterization campaign.

2.2 Bayesian calibration and experimental design

Bayesian calibration represents parameter uncertainty conditional on a forward model, likelihood, and prior [17]. Computer-experiment literature distinguishes emulator un-

certainty, observation error, calibration parameters, and structural discrepancy [18, 19]. Validation frameworks consequently compare predictions with observations rather than treating posterior concentration as proof of physical validity [20]. Ignoring discrepancy can bias inferred physical parameters and produce overconfident extrapolation [21].

Bayesian experimental design formalizes the value of a proposed observation through expected utility. Lindley’s information criterion [22], foundational reviews [1, 2], and classical optimum-design methods [23] motivate the EIG criterion used here. Estimation remains challenging because the predictive density is generally unavailable; relevant strategies include Laplace and MCMC approximations [24], nested Monte Carlo [25], stochastic simulation-based optimization [26], and variational bounds [27]. Adaptive-policy amortization offers a scalable direction for longer experimental sequences [28], but the present work deliberately uses a transparent one-step ranking over a finite candidate library.

III. MAGNETIC-CORE MODELS

3.1 Core loss

For sinusoidal excitation within a specified operating window, the Steinmetz model is

$$P_v(f, B_{pk}) = k f^\alpha B_{pk}^\beta, \quad (1)$$

where P_v is volumetric loss and (k, α, β) are inferred parameters [3]. The units of k depend on the units used for P_v , f , and B_{pk} . Equation (1) is not temperature dependent in the present implementation. Measured-data analysis therefore uses a specified temperature subset; extrapolation across temperature is out of scope. More general waveform-dependent loss equations remain important for converter application studies [6, 9].

3.2 Complex permeability

We adopt the $e^{j2\pi ft}$ time convention and write $\mu_r = \mu' - j\mu''$, so the reported loss component is $\mu'' = -\text{Im}\mu_r \geq 0$ for a passive response. The relative complex permeability is represented with a single Cole–Cole relaxation [13],

$$\mu_r(f) = 1 + \frac{\mu_s - 1}{1 + (jf/f_{rel})^{1-a_{cc}}}, \quad (2)$$

where μ_s is low-frequency permeability, f_{rel} is a relaxation frequency, and a_{cc} broadens the relaxation. Storage and loss components are evaluated separately because a model can fit μ' while failing to represent μ'' . A one-pole model is a deliberate low-order approximation, not a complete ferrite loss model.

Equation (2) fixes the high-frequency asymptote to unity and absorbs the angular-frequency convention into f_{rel} . The real and loss components follow from the same complex denominator, which prevents independent tuning of μ' and μ'' . For an optional inductance channel, the real component is mapped through $L_m = \mu_0 \mu' N^2 A_e / \ell_e$ using specified geometry [11, 12]. Geometry is never inferred jointly with material parameters in the present study.

3.3 Validity window

Both forward laws are isothermal and evaluated only inside specified data ranges. A data point is defined by channel, frequency, peak flux density, and temperature. Core-loss

points require positive B_{pk} ; permeability points use small-signal records. The likelihood rejects mixed temperature cohorts. Measured permeability uses material-specific cohorts of $25 \pm 0.75^\circ\text{C}$ for N87 and $30 \pm 0.75^\circ\text{C}$ for N95. Measured core loss uses $25 \pm 0.5^\circ\text{C}$ for N49, N87, and N95, and $100 \pm 0.5^\circ\text{C}$ for 3C95. These restrictions prevent an apparently precise posterior from silently extrapolating across temperature, waveform, or excitation regime.

IV. BAYESIAN CALIBRATION AND DESIGN

Let $\theta = (\log k, \alpha, \beta, \log \mu_s, \log f_{rel}, a_{cc})$ denote the active transformed parameters. Log coordinates enforce positivity of scale parameters while retaining direct coordinates for the exponents. For observation y_o at design d_o , the likelihood is

$$p(\mathbf{y} | \theta, \mathbf{d}) = \prod_o \mathcal{N}(y_o; F_o(\theta, d_o), \sigma_o^2), \quad (3)$$

with fixed relative scales. Synthetic scales are $3\mu'$, 5relative scale as a weighting model. The posterior is

$$p(\theta | \mathbf{y}, \mathbf{d}) \propto p(\mathbf{y} | \theta, \mathbf{d}) p(\theta). \quad (4)$$

Priors are defined directly in the transformed space used by the sampler, so no inverse-transform Jacobian is added to this active-coordinate density. Bounds encode physically admissible numerical domains, not empirical proof that the chosen model is correct.

4.1 Posterior computation and diagnostics

Posterior sampling uses the affine-invariant ensemble construction [29] implemented by `emcee` [30]. Each ensemble retains 5000 steps after 1000 warm-up steps for each of 48 walkers. Initial states are perturbations of the prior center and both initialization and proposal streams are deterministically seeded. A fit is accepted only with finite integrated autocorrelation estimates, at least 50 retained steps per autocorrelation time, effective sample size at least 400, mean acceptance between 0.20 and 0.60, and finite log probability for every retained sample.

These checks follow the principle of comparing Monte Carlo variation with between-sequence behavior [31] and modern effective-sample-size practice [32], but interacting ensemble walkers are not treated as independent chains. Accordingly, the paper does not report a conventional $\hat{R}$ for the ensemble. Diagnostic failure invalidates a measured fit for quantitative publication rather than being repaired by discarding an unfavorable seed.

The local Fisher approximation [33] is

$$\mathcal{J}_{ij} = \sum_o \frac{1}{\sigma_o^2} \frac{\partial F_o}{\partial \theta_i} \frac{\partial F_o}{\partial \theta_j}. \quad (5)$$

Its spectrum is computed over the exact active vector and with reported scaling. A large condition number indicates locally sloppy combinations but does not by itself establish global non-identifiability [34]. Geometric analyses of sloppy models likewise interpret broad eigenvalue spectra as anisotropic distinguishability, not as a universal parameter ranking [35]. The spectrum is therefore reported together with its dimension and scaling convention.

Let $\mathcal{D}_t = (\mathbf{y}_t, \mathbf{d}_t)$ denote the observations available before acquisition step t . For candidate d , the one-step EIG is the

conditional mutual information [22, 1]

$$\text{EIG}_t(d) = \mathbb{E}_{\theta \sim p(\theta | \mathcal{D}_t)} [\log p(y | \theta, d) - \log p(y | d, \mathcal{D}_t)], \quad (6)$$

where

$$p(y | d, \mathcal{D}_t) = \int p(y | \vartheta, d) p(\vartheta | \mathcal{D}_t) d\vartheta. \quad (7)$$

The nested Monte Carlo estimator draws 1200 outer parameter–observation pairs and uses 400 inner posterior draws to approximate the posterior-predictive density [24, 25]. Each score is repeated 40 times; the mean, standard error, 95% Monte Carlo interval, and top-rank frequency are retained. Candidate streams are seeded from a stable key comprising channel, frequency, flux density, and temperature, so list permutation cannot change a design’s draws. We evaluate two distinct objectives: raw EIG for the measurement-count endpoint and EIG divided by a prespecified modeled acquisition cost for the total-cost endpoint.

4.2 Paired sequential policy

After the same two initial observations— P_v at 30 kHz, 0.05 T, and 25 °C, and μ' at 10 kHz and 25 °C—each policy selects one unrevealed candidate, receives its pre-generated noisy outcome, refits the posterior, and checks the common precision gate. Raw EIG maximizes estimated information, EIG/cost maximizes information per modeled cost, and the comparator follows a deterministic fixed channel-balanced grid traversal. It is not uniform random sampling, a space-filling design, or a D-optimal design. All policies share the candidate library, latent truth, candidate-indexed noisy outcomes, maximum budget, and stopping rule. The candidate library is isothermal, so it contains no temperature-only duplicates of the implemented forward laws. The numerical settings are summarized in Table I.

The stopping statistic is the half-width of the central 90% posterior interval for the noise-free latent mean response divided by the absolute posterior median. Observation noise is not added to this gate. The two fixed targets are $P_v(100\text{kHz}, 0.1\text{T}, 25^\circ\text{C})$ at 8% and $L_m(100\text{kHz}, 25^\circ\text{C})$ at 5%. This is a local precision criterion: it checks neither truth proximity, six-parameter recovery, other channels, held-out performance, nor model adequacy. A policy that misses either threshold is recorded as a failure at the maximum budget.

V. COMPUTATIONAL REPRODUCIBILITY

All experiments use fixed software versions, specified model and inference settings, explicit data selections, and deterministic random seeds. Results are retained at the individual-seed level and are accepted for aggregation only when the prespecified convergence, validity, and completeness criteria are satisfied.

The analysis checks all required seed and material cases before forming aggregate summaries. Records with nonfinite values, failed convergence, or boundary-active parameters are excluded according to criteria specified before evaluation. Numerical statements, tables, and figures are generated programmatically from the same complete set of accepted records, avoiding manual transcription or selective replacement of outcomes.

TABLE I: COMPUTATIONAL SETTINGS USED FOR THE REPORTED EXPERIMENTS.

Item	Setting
Ensemble sampler	48 walkers; 1000 warm-up + 5000 retained
Convergence criteria	ESS $\geq$ 400; steps/ $\tau \geq$ 50
Acceptance criterion	0.20–0.60
Nested EIG	1200 outer; 400 inner; 40 score replicates
Sequential budget	at most 25 measurements
Acquisition endpoints	raw EIG/count; EIG/cost/modeled cost
Temperature cohorts	material and channel specific, as stated in text
Precision gate	latent mean response: 8% P_v ; 5% L_m

VI. EVALUATION PROTOCOL

Five fixed seeds are used for matched-model recovery. A disjoint, prespecified set of 30 seeds is used for the paired acquisition comparison only after the nested-Monte-Carlo setting passes a separate convergence study. That study uses prefix-nested common random numbers and a deterministic paired bootstrap of the replicate-mean selector; it requires stable candidate and reference winners, rank correlation, score-interval overlap, low reference regret, and downstream endpoint agreement. The policies share initial observations, candidate library, realized candidate outcomes, noise policy, budget, and stopping rule. Results include per-seed counts, modeled costs, and failures, not only an aggregate ratio. The stopping rule concerns posterior precision of two latent mean responses and is not an accuracy or coverage guarantee.

For recovery, observations are generated from the same six-parameter forward family used in inference. A parameter vector is drawn independently for each seed before observation noise is generated. The reported error is the absolute relative difference between posterior median and the generating value. The interval statistic counts whether that value falls within the equal-tailed 90% model-conditional posterior credible interval. With only five seeds, the count is a descriptive implementation check, not a coverage experiment. The Fisher matrix is evaluated in the exact transformed coordinate system used by sampling.

The measured evaluation uses public complex-permeability curves for specified N95, N87, and 3C95 source/material pairs and datasheet core-loss curves for N49, N87, N95, and 3C95 [16]. Input-data versions, temperature cohorts, unit conversions, and range filters were fixed before inference. Storage/loss permeability errors, core-loss residuals, boundary activity, and operating-window coverage are reported separately. Records with failed convergence or active boundary flags are retained for diagnostic review but excluded from aggregate numerical claims.

Core-loss adequacy is summarized by in-sample relative root-mean-square error (RRMSE) within each temperature cohort. Permeability adequacy is evaluated separately for storage and loss components because combining them can conceal a poor μ'' fit behind the larger magnitude of μ' . These metrics assess the selected low-order model on observed records; they are neither held-out prediction errors nor component-level thermal validation.

TABLE II: MATCHED-MODEL RECOVERY AND INCLUSION OF THE GENERATING VALUE IN EQUAL-TAILED 90% POSTERIOR CREDIBLE INTERVALS (FIVE SYNTHETIC SEEDS).

Parameter	Median error (%)	Truth included
k	5.85	4/5
α	0.27	4/5
β	0.36	5/5
μ_s	0.38	5/5
f_{rel}	0.87	5/5
a_{cc}	0.62	5/5

VII. RESULTS

7.1 Identifiability and synthetic recovery

The synthetic 6×6 Fisher matrix was full rank. As shown in Fig. 1(a), its eigenvalues span 1.24×10^2 to 2.91×10^6 , giving a full-matrix condition number of 2.35×10^4 . The broad spectrum indicates strong local sensitivity anisotropy in the active transformed coordinates; it does not establish global identifiability. Fig. 1(b) shows the absolute posterior-median errors for each parameter and seed. Across five prior-predictive matched-model data sets, the per-parameter medians range from 0.27% to 5.85%. The known values fell within 28/30 equal-tailed 90% model-conditional posterior credible intervals. This observed inclusion count is descriptive, not a coverage estimate. Table II reports the same matched-model recovery check.

7.2 Paired synthetic acquisition

Fig. 2(a) makes the paired outcome structure explicit. Raw EIG reached the two-target latent mean-response precision gate with 4–5 measurements, while the fixed channel-balanced grid traversal required 9–9. Raw EIG used fewer measurements in every paired run, with a mean reduction of 44.8%. These results evaluate selection efficiency for the count endpoint in this 30-seed matched-model experiment. The mean paired count difference has a bootstrap 95% interval of 4.00–4.10 measurements; the paired win rate is 100.0% and the raw-EIG gate-failure rate is 0.0%. For the separate modeled-cost endpoint, EIG/cost gave a descriptive mean paired reduction of 34.4%; neither policy comparison had a gate failure. The cost units are prespecified rather than measured laboratory time.

7.3 Measured-data model adequacy

Fig. 2(b) compares the in-sample relative root-mean-square errors (RRMSEs) of the accepted measured-data fits. For LEA Material Database permeability records that satisfied the convergence criteria, RRMSEs were 9.33%/52.42% for μ'/μ'' on N87 and 6.89%/36.77% on N95. The larger μ'' errors expose discrepancy in the one-pole model. N87 and 3C95 MagNet records failed convergence or boundary gates and were excluded from numerical claims [15]. Measured Steinmetz fits produced in-sample RRMSEs of 8.79% (N87), 12.55% (N49), 13.40% (3C95), and 18.21% (N95) within their specified temperature cohorts. These are fit residuals, not held-out errors.

All numerical results in this section, including Figs. 1 and 2, use the same prespecified models, data filters, random

seeds, and evaluation criteria.

VIII. DISCUSSION AND LIMITATIONS

8.1 Interpretation of the evidence

The framework separates whether available measurements constrain parameters within a chosen model from whether that model represents measured material behavior. Fisher information and posterior width address the former locally or conditionally; reported in-sample residuals expose, but do not fully quantify, the latter. Held-out predictive validation remains future work.

The synthetic results support implementation-level conclusions. Full local rank shows that every active coordinate affects the specified synthetic experiment, while the broad spectrum shows that those effects occur at very different scales. The recovery table then checks the nonlinear posterior under matched truth. Neither result implies that the same coordinates are uniquely physical when the material response lies outside the six-parameter family. This distinction matters for converter models because several parameter sets can produce similar loss or permeability over a narrow operating window yet diverge when frequency, flux density, or temperature changes.

The paired acquisition result shows that raw EIG exploited posterior structure in the matched problem: it reached the two-target precision gate with fewer measurements for every reported seed under shared candidate outcomes. The result means only that fewer measurements were needed to narrow two specified latent mean-response intervals. It does not establish accurate recovery of the magnetic component, global predictive accuracy, or robustness to model misspecification.

8.2 Implications for converter modeling

For a converter workflow, the posterior is most useful as a conditional input to loss, inductance, and sensitivity calculations over the specified operating region. Posterior draws can be propagated through a component model instead of substituting one nominal coefficient set, allowing a designer to determine whether parameter uncertainty materially affects thermal or magnetic design margins. The Fisher spectrum can motivate testing whether a new channel or a wider, physically controlled excitation range would constrain a weak direction more effectively than densifying the original sweep [1, 23].

Measured residuals define a separate model-adequacy assessment. The accepted core-loss fits show the error remaining after estimating Steinmetz coefficients inside one temperature cohort. The much larger loss-component permeability residuals show that a single Cole–Cole pole cannot be treated as a broadband physical description for these records. A component calculation that is sensitive to μ'' should therefore retain an explicit discrepancy allowance or adopt a higher-order permeability law before relying on posterior parameter width. Conversely, a reasonable μ' residual does not validate μ'' , because the two components occupy different numerical scales and correspond to different design consequences.

The common evaluation protocol safeguards this interpretation by preventing combination of a favorable recovery run, a figure from another seed set, and a measured fit produced under different filters. Fixed model settings, data

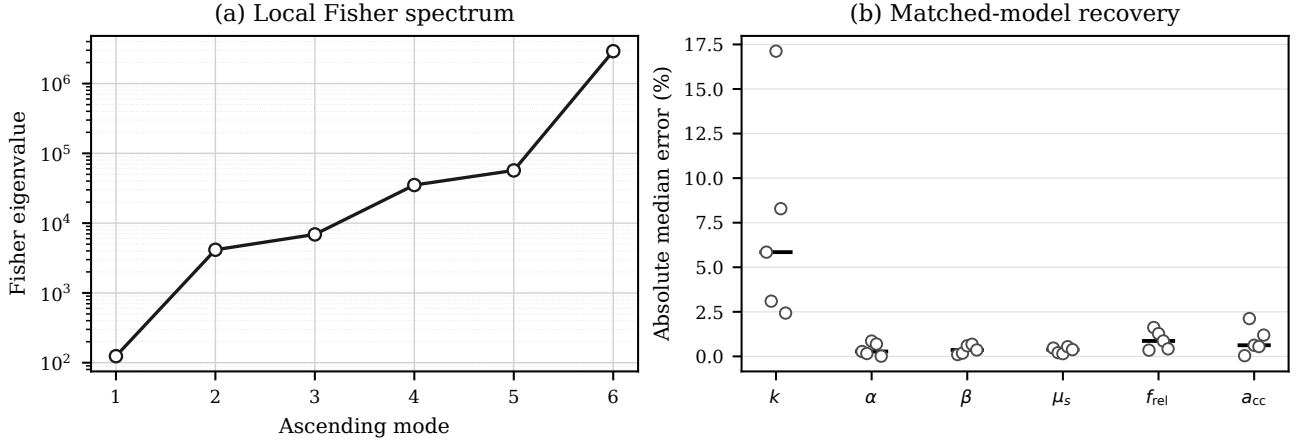


Fig. 1. Synthetic local identifiability and matched-model recovery. (a) Fisher eigenvalues in ascending order. (b) Absolute posterior-median errors across five seeds; circles are horizontally offset for visibility and horizontal ticks mark medians.

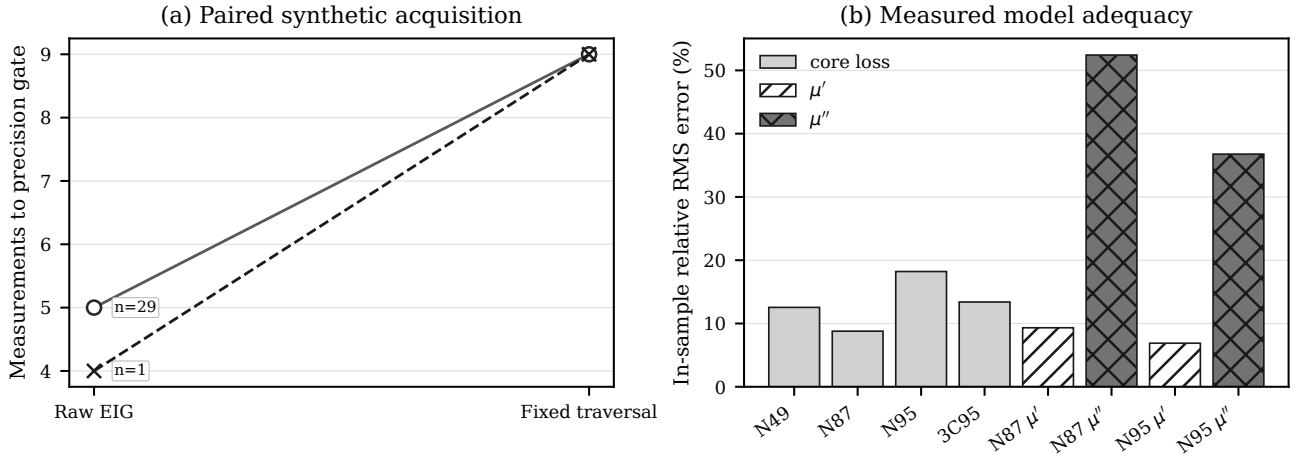


Fig. 2. Paired synthetic acquisition and measured-data model adequacy. (a) Measurements required to reach the two-target latent mean-response precision gate; each label lists seeds sharing the same paired outcome, and line styles expose coincident endpoints. (b) Measured-data in-sample RRMSE; fill and hatch distinguish core loss, μ' , and μ'' . Excluded MagNet fits are not plotted.

selections, and random seeds also make it possible to distinguish sampling variation from changes to the computational experiment.

8.3 Threats to validity

Internal validity is limited by five synthetic recovery seeds and 30 paired acquisition seeds. Pairing controls differences in candidate outcomes between policies. A prespecified convergence study compares candidate ranks and downstream endpoints against a higher-budget reference; finite nested Monte Carlo variance nevertheless remains. Synthetic recovery is matched-model, and the observed interval-inclusion count is too small to establish calibrated frequentist coverage.

Construct validity is limited by the low-order forward models and available data. The Steinmetz relation is temperature independent, and one Cole–Cole pole does not capture all ferrite relaxation, resonance, or loss processes [14]. Measured errors are in-sample. Public curves may also contain digitization, instrument, preprocessing, and source-

specific errors not represented by the likelihood. Excluding convergence-invalid or boundary-active records prevents unsupported aggregate claims but restricts the population represented by the accepted subset.

External validity is confined to the specified channels, temperature cohorts, excitation ranges, materials, and cost model. The measured-data EIG ranking is a model-conditional acquisition suggestion, not a validated optimal laboratory plan, particularly given the observed μ'' discrepancy. Measurement-count reductions must not be interpreted as laboratory-time savings because setup, settling, calibration, rejection, and batch costs were not measured. The results do not replace held-out validation, controlled multi-lot characterization, switching-waveform experiments, or component qualification. Those extensions require new experimental evidence rather than broader interpretation of the present results.

IX. CONCLUSION

Bayesian calibration recovered the six active Steinmetz and Cole–Cole parameters with per-parameter median absolute relative errors of 0.27%–5.85% across five matched-model synthetic data sets. The known values fell within 28/30 equal-tailed 90% model-conditional posterior credible intervals. Under paired candidate outcomes and a common stopping rule, raw EIG reached the two-target latent mean-response precision gate with 4–5 measurements, compared with 9–9 for the fixed channel-balanced grid traversal, a descriptive mean reduction of 44.8% across 30 seeds. The paired count-difference bootstrap interval was 4.00–4.10 measurements, with 100.0% paired wins and a 0.0% raw-EIG gate-failure rate. The accepted measured records produced substantially larger RRMSEs for μ'' than for μ' , showing that posterior concentration cannot compensate for an inadequate permeability law. The workflow links inference, local identifiability, measurement selection, model-adequacy checks, and traceable evidence within the tested models, public data records, and operating ranges.

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